



SCHOOL OF COMPUTER TECHNOLOGY

AASD 4007
Design Thinking for A.I. Solutions
Session 4

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Session 4



- Debriefing In-class Exercise 2
- Design Thinking Exercises: (In-class)
 - AI Essentials: Data
 - Prioritization
 - To-Be-Scenario Map (A4 revisited)
 - Hills
 - Debriefing to the class
- Prototyping
- A5: Tasks & Low Fidelity Prototype (Homework)
- Term project

Debriefing In-class Exercise 2



- Empathy map
- As-Is scenario map
- Needs statements
- AI intents
- AI big idea vignettes
- **Debriefing to the class**

Design Thinking Exercises



In-class Exercise 3 = 10 % of grade

Mode: in group

Submission: on Brightspace as a MURAL link

Time: 100 min

- AI Essentials: Data
- Prioritization
- To-Be-Scenario Map (A4 revisited)
- Hills
- Debriefing to the class

AI Essentials: Data



<https://www.ibm.com/design/thinking/page/toolkit/activity/ai-essentials-data>

1. Write down all the data you have, place them in Public, Private, User categories
2. Write down all the data you want to have. Discuss if they are available
3. Vote on the most critical data sources. Define minimum data sources you need to meet your AI intents

GROUP ACTIVITY (20 min)

AI Essentials: Data

Sub-space: Transit

	AI DATA	PUBLIC	PRIVATE	USER
HAVE	Daily Public Transport Passenger Journeys by Service Type in Ontario Covid-19 Data Explorer with data such as deaths, cases, vaccination, mobility risk, policy responses COVID-19 daily epidemiology update COVID-19 case surveillance dataset COVID-19 worldwide vaccination progress TTC dataset about the passenger travel details Face Mask Detection Video Dataset Face masks detection Data to Insights for a Better Canada Symptom and body temperature distributions at time of triage among healthcare workers			
WANT/NEED	Air quality standard for public transit Videos of bus passengers with symptoms labels Collect live passenger data Distant detection data Social distancing of people within the set radius Data of people who are still not vaccinated DNA record Passenger vaccination record			
NICE TO HAVE	Data about effectiveness of removing seats on covid transmission Effective social distance on public transit Face mask data (whether the person is wearing a mask or not) Photos of passengers with symptoms label Public transit cases per day per subway/bus/route Passenger health record			

Prioritization



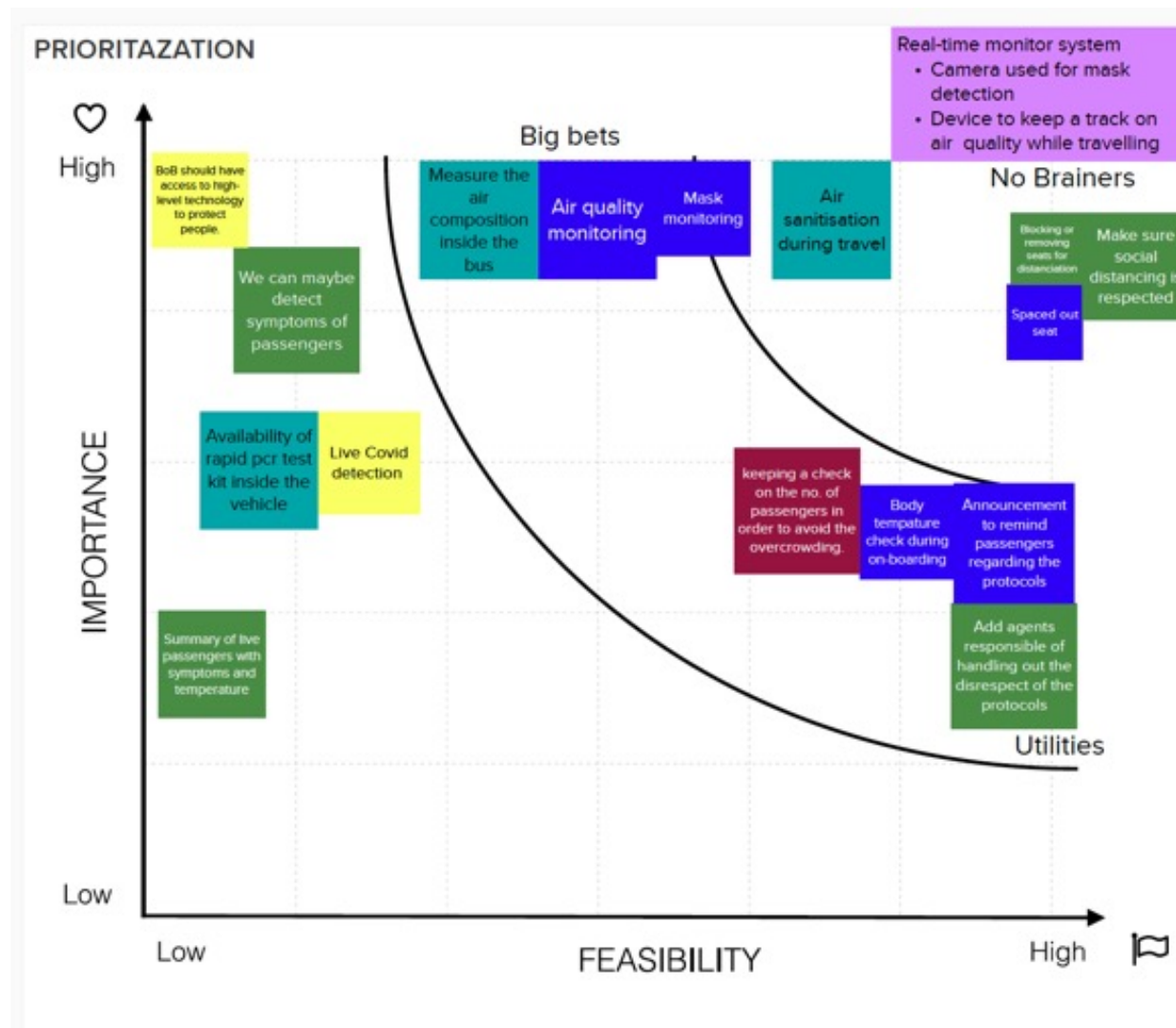
<https://www.ibm.com/design/thinking/page/toolkit/activity/prioritization>

VIDEO [01:14]

1. Use Big Idea Vignettes & plot them on grid below
2. Discuss their placements & move ideas if necessary
3. Label **No Brainers, Utilities, Big Bets** & discuss details

GROUP ACTIVITY (20 min)

Prioritization



To-Be-Scenario Map



<https://www.ibm.com/design/thinking/page/toolkit/activity/to-be-scenario-map>

1. Imagine your solution exists. Describe user experience with your solution
2. Cluster and order by time. Label different stages of the experience
3. Identify highs & lows. Compare to As-Is-Scenario & identify changes your solution can bring to your user

GROUP ACTIVITY (30 min)

To-Be-Scenario Map



Outline the steps a user will take to complete a task

- Centered around ONE task
- i.e. “Booking a hotel for a business trip”, “Going to the airport and getting on a plane”, “Booking a vaccine appointment”
- If your scenario is too broad, try breaking it down into smaller aspects of the experience

To-Be-Scenario Map

Real-time monitor system
• Mask detection and track on air quality while travelling

Steps

Check the bus before starting

Welcoming passengers on board

Driving the bus

Return to the terminal

Doing



Checks the driver shield and masks on

Makes sure that his window is slightly open to get ventilation

Makes sure that the device installed is working properly by testing it

Turn on real-time counting monitor system

Turn on cameras

Check if passengers are wearing a face mask before entering the bus

check if passengers are carrying a vaccination certificate

Make sure everyone is following the rules

Opening doors

Monitor real-time data on the bus to ensure everyone is doing their part

disinfecting buses with ultrasonic sanitizer

During changing shifts, follow social distancing with the colleague

Thinking



The bus will be safe for everyone with the monitoring system

How many trips that he should be doing that day

Everyone will be safe if rules are followed

Able to start working

where to get vaccinated

Is everybody going to respect sanitary measures?

Is the device installed working properly

What if the device goes down due to technical issue

Everyone is safe based on the real-time monitor system on the bus

Is the driver for the next shift available?

Everyone followed the rules during the ride

The real-time monitoring system was working fine and great

Feeling



Relieved because technology takes care of safety

Confidence that the system will work and provide safety for everyone

Feeling safe that the passengers now have to follow the mask regulation strictly

Happy to serve

Stressed if the system goes down

Safe and protect with the monitoring system

feeling of new normal and things becoming easier with the help of technology

Relief and relax

<https://www.ibm.com/design/thinking/page/toolkit/activity/writing-hills>

VIDEO [01:21]

1. Individually diverge on maximum number of ideas
2. Combine ideas into sentences to describe your intention for the user
3. Edit, refine, & iterate. Get feedback on your hills and update further

GROUP ACTIVITY (30 min)

The row of six images illustrates various ways to organize sticky notes on a whiteboard:

- Image 1:** A mind map centered on a drawing of a person's head with the name "LIZ" written above it. Branches extend outwards with sticky notes containing text like "I have a job", "I need help", "what do I want?", "I'm a realist", "I'm a dreamer", and "LEADS".
- Image 2:** A grid layout with sticky notes organized into columns under headers "WHITE", "GREEN", and "PINK".
- Image 3:** A grid layout with sticky notes organized into columns under headers "WHITE", "GREEN", and "PINK".
- Image 4:** A grid layout with sticky notes organized into columns under headers "WHITE", "GREEN", and "PINK".
- Image 5:** A flowchart or process diagram with sticky notes arranged in a sequence, connected by arrows. It includes a "START" circle and a "FINISH" circle.
- Image 6:** A grid layout with sticky notes organized into columns under headers "WHITE", "GREEN", and "PINK".

Needs Statements

Why do we think these ideas are so important and impactful for our user? What does our user actually seek?

HILLS

How to
build a Hill

IBM A THOMSON COMPANY

Who

Specific user or class of users.

What

Specific user enablement.

Wow

Specific and differentiating value to the user.

HILLS

Yellow
flags in
Hill-writing



Improve
Enhance
Better
Accurate
Easy-to-use

Quickly
Intuitively
In (#) clicks
Next-generation
Best-in-class

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HILLS

A Good Hill

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Within selected product categories,
requestors can find product matches
for their search queries using natural,
English-language conversation.

Who What Wow

HILLS

A Good Hill

IBM A GEORGE BROWN COLLEGE COURSE

A procurement manager can change buying strategies based on global market conditions, as fast as social media reveals them."

Who What Wow

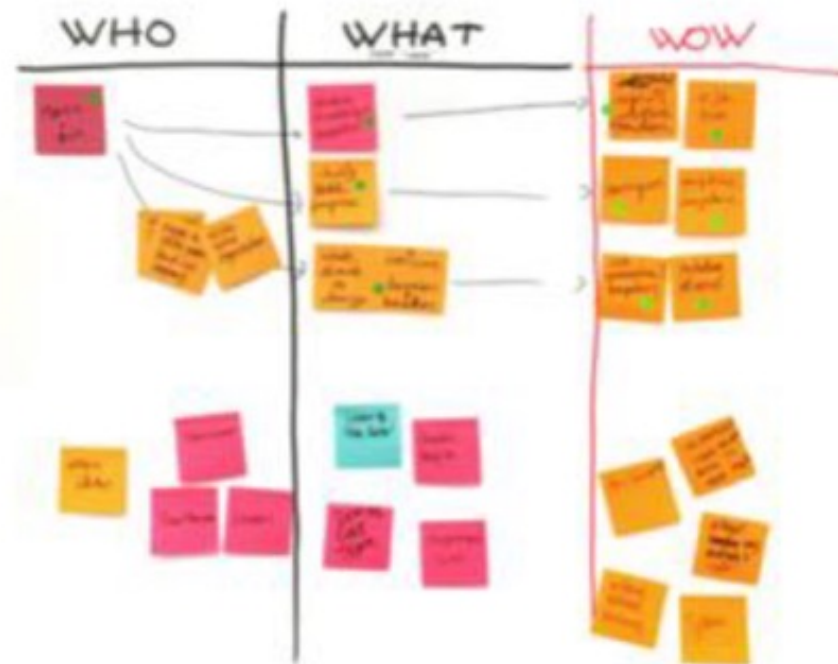
Hills

HILLS

Exercise: Write Hills

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A 1974 IBM Corporation IBM Study



HILLS

Instructions:

1. Ideate on Whos and Whats

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Whos

Focus on specific users or groups of users

- A financial analyst...
- An HR manager...

Whats

Focus on user enablement

Are not Hows

Often start with:

- Can...
- Will...
- Is able to...



HILLS

Instructions:

2. Converge

IBM

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Whos

Focus on specific users or groups of users

- A financial analyst...
- An HR manager...

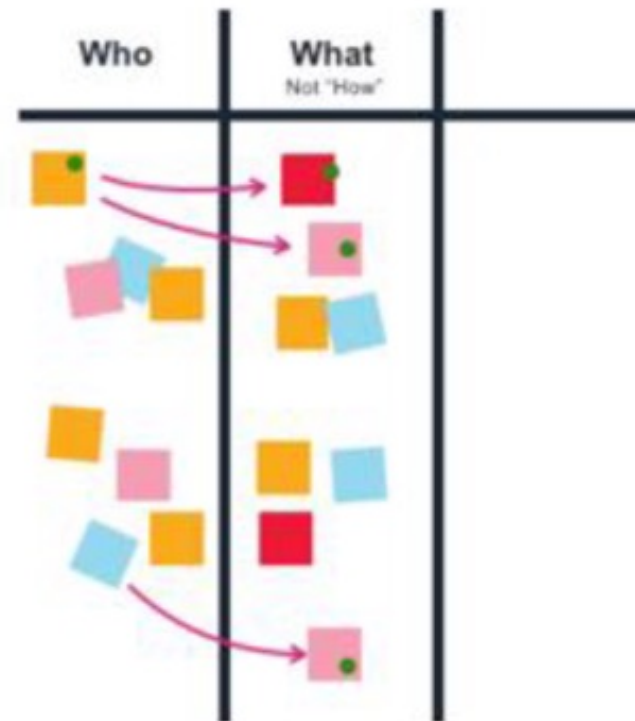
Whats

Focus on user enablement

Are not Hows

Often start with:

- Can...
- Will...
- Is able to...



3. Add Wows

© 2006-2007 Blackwell Publishing Ltd, *J. Oral Maxillofac. Surg.* 65: 1035-1040

- Time
- Resources
- Skill Level
- Processes
- Location
- Dependencies



HILLS

Instructions:

4. Converge again

IBM

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Wows

Are from the perspective of your user

Are market differentiating

Are measurable

Often focus on:

- Time
- Resources
- Skill Level
- Processes
- Location
- Dependencies



HILLS

Who

Bus driver

TTC
passengers

The TTC

Employees

What

Make sure
passengers
are following
rules

Feel safe at
work

can feel safe
while taking
public transit

Limit the
spread of
COVID by
using public
transit

provides new
solutions to
the covid
problem

Can make
sure
passengers
are doing
their part

Wow

can monitor
the
passengers
on the bus in
real-time

with the real-
time monitor
system

by knowing
everyone on
the bus
followed rules
in real-time

visual system
that is going to
provide you
details about the
air quality of the
bus

with the real-
time monitor
system on
their buses

One stop
solution for
covid related
issues in the
public transport

Bus driver can make sure passengers are following rules and feel safe at work with the real-time monitoring system.

TTC passengers can feel safe while taking public transit by knowing everyone on the bus followed rules in real-time.

The TTC can help limit the spread of COVID in the public transport system by using the monitoring system.

Prototyping



- Visual representation of an idea before developing it, an early sample, model, scale-down version or release of a product built to test a concept or process
 - Ex. cardboard, paper, plastic
 - Allows for getting feedback and testing it with team members/target users to improve and revise the idea before creating it
 - Helps answer:
 - Does the product solve the user problem?
 - Does it score high on user desirability?
 - Is it user-friendly?

Three type of prototypes:

- Low Fi
- Medium Fi
- High Fi

Fidelity refers to the detailing and the functionality of the prototype

Prototyping

Prototypes



Early Design

Late Design

Development



Costs of Marketing Changes:

\$1

\$10

\$100

Prototype Fidelity

Faithfulness of prototype appearance and performance to final product.

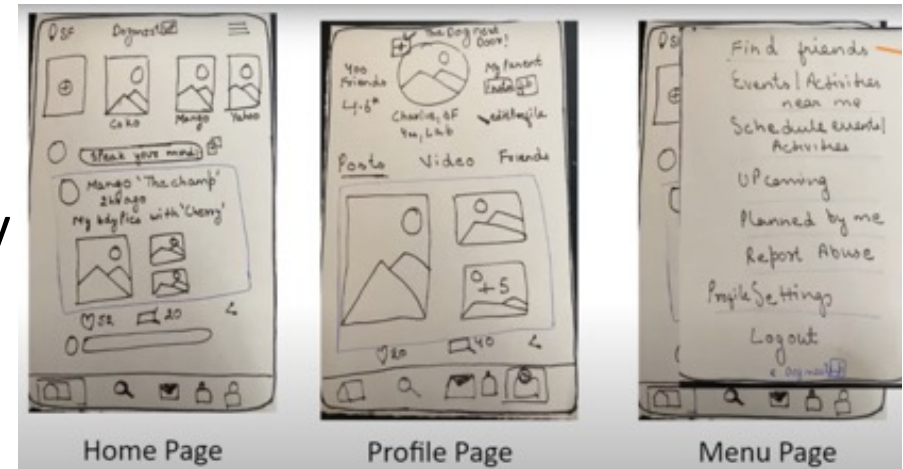
- Low fidelity: prototype doesn't look much like final product, operation of the prototype may be simulated and slower.
- High fidelity: prototype looks and operates like the real product.



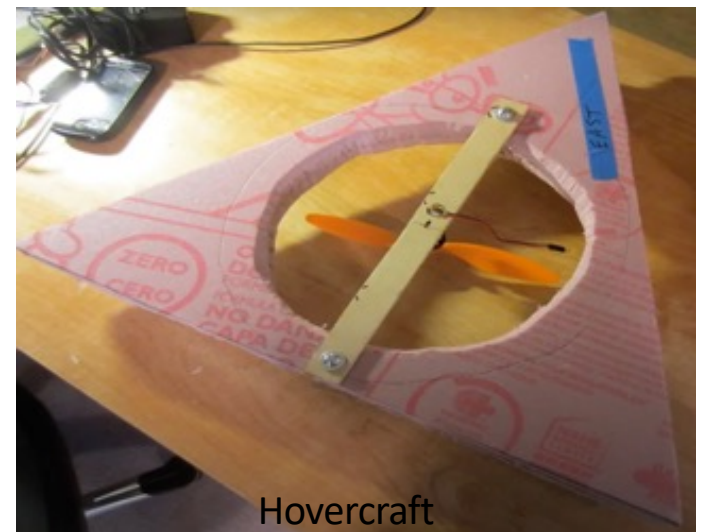
Prototyping

Low Fidelity Prototype

- For early design phases
- Can make changes quickly & cheaply
- For example:
 - Paper sketch
 - Paper prototype
 - Cut-outs
- Paper prototype
 - Different sketches of the user interface
 - Simulate interactivity by linking/moving paper



Mobile application



Hovercraft

Paper Prototyping Goal

Maximum User Feedback
for
Minimum Effort

Prototyping

Paper Prototyping Technology



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Prototype – Palm Pilot

<https://vimeo.com/552613636>



Figure 10: Reproduction of 3D "Sketch" of the PalmPilot
Very early on, Jeff Hawkins carved up a piece of wood to get a sense of the size that the device must be. It needed to fit into a shirt pocket, as well as sit comfortably in the palm of the hand.
(Photo: Elizabeth Russ)

Prototype – Shopping Cart



IDEO's Shopping Cart Redesign

ABC's Nightline Program, 1999

<https://www.youtube.com/watch?v=taJOV-YCieI>

Prototyping

Prototype – TiVo Remotes

The New York Times

February 18, 2004



Four Delivers for The New York Times

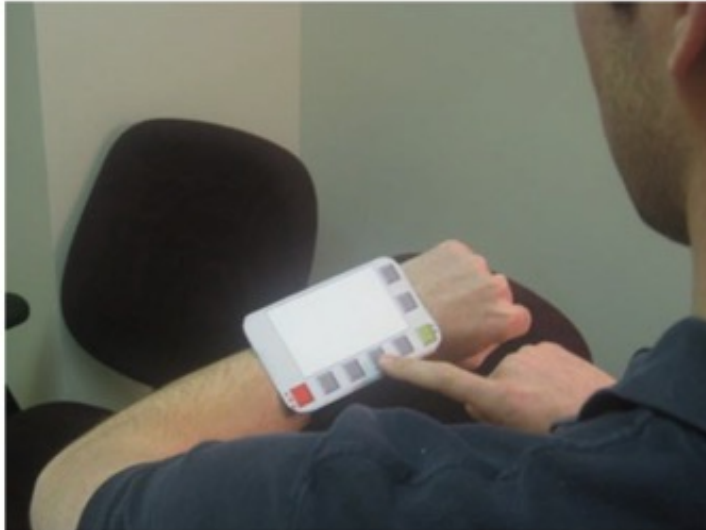
Then they tested the feel and look of various three-dimensional models carved from rigid foam.



Prototype – Medical Equipment



Prototype – Physical Device



<http://www.sketchingtomorrow.com/wp-content/uploads/2010/06/cardboard.jpg>

IDEO Service Prototyping Postpartum Patient Experience

<https://www.youtube.com/watch?v=Szw6KITMUT4>

HOW MIGHT WE IMPROVE THE POSTPARTUM PATIENT EXPERIENCE?

Kaiser Permanente



Prototyping

Prototype – Kiosk



www.nngroup.com

Assignment 5: Homework



Tasks & Low Fidelity Prototype

Term Project



- Presentations (in-person), slides are due Apr 17th by 5pm
Upload your files through Brightspace: Term Project>Project Presentations
- Reports are due Apr 19th by midnight
Upload your files through Brightspace: Term Project>Project Reports